Automated Network Request Management in ServiceNow

PROJECT REPORT

Project Name	Automated Network Request Management in ServiceNow
Phase	Full Project — All Phases
Author	Darapu Subhakar
Team ID	DARAPUSUBHAKAR-NRM-01
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Platform / Domain	ServiceNow / IT Service Management (ITSM)
Academic Year	2025 – 2026
Project Type	Academic / Industry Simulation Project
Tools Used	ServiceNow Flow Designer, Service Catalog, Workflow Editor, Form Designer

1. Introduction

1.1 Project Overview

In modern enterprises, managing network-related service requests through manual or fragmented systems generates significant operational inefficiencies. IT teams frequently receive requests via emails, phone calls, spreadsheets, or informal communication channels, resulting in delayed responses, poor audit trails, lack of real-time visibility, and difficulty tracking request status. Network provisioning, access grants, firewall rule changes, VPN setup, and bandwidth upgrades all demand a structured, traceable, and automated approach to be handled reliably.

To address these challenges, this project presents the **Automated Network Request Management System** — a comprehensive solution developed on the ServiceNow platform. The system digitizes and automates the full lifecycle of network service requests, from submission and categorization through approval, fulfillment, and closure, within a centralized and auditable framework. Role-based access control ensures requesters, approvers, network engineers, and administrators each interact with the system according to their responsibilities. Workflow automation through Flow Designer routes requests through predefined approval and fulfillment chains, triggers email notifications at each stage, and creates assignment tasks for the appropriate network teams.

1.2 Problem Statement

Organizations operating complex network infrastructures face recurring challenges when managing network service requests through traditional, unstructured methods. The absence of a centralized intake system results in the following critical pain points:

- Requests submitted via email or phone lack traceability and are frequently lost or overlooked.
- No standardized intake process leads to incomplete or inconsistent request data.
- Informal approval workflows create compliance risks and audit failures.
- Network engineers lack a unified task queue, causing confusion over ownership and priority.
- Requesters have no real-time visibility into the status of their submissions.
- SLA monitoring and escalation are entirely manual, resulting in frequent breaches.
- Reporting and analytics on network request trends are extremely difficult to generate.

1.3 Objectives

- Design and deploy a structured Service Catalog with categorized network request items.
- Build dynamic, intelligent forms that capture all required data for each request type.
- Implement role-based access control ensuring appropriate visibility and actions for each user group.
- Configure automated multi-level approval workflows aligned with organizational hierarchy and compliance requirements.
- Integrate email notifications that keep all stakeholders informed at every stage.
- Establish automated task assignment for network fulfillment teams.
- Enable SLA tracking with automated escalation mechanisms.
- Validate the system end-to-end through rigorous testing including User Acceptance Testing (UAT).

- Deploy the solution in a production environment with proper training and support processes.

1.4 Scope of the Project

The scope of this project encompasses the full design, development, testing, and deployment of the automated network request management system within the ServiceNow platform, covering:

- Service Catalog creation with categorized network request types.
- Dynamic form setup with field-level validation and conditional logic.
- Multi-level approval workflow configuration using Flow Designer.
- Automated email notifications and comprehensive audit logging.
- Task creation and assignment to network fulfillment groups.
- SLA definition, monitoring, and automated escalation for each request category.
- End-to-end testing including unit testing, integration testing, and UAT.
- Production deployment with training materials and post-deployment support.

2. Literature Review

2.1 IT Service Management and ITSM Frameworks

IT Service Management (ITSM) refers to the set of processes, policies, and procedures organizations use to design, deliver, manage, and improve IT services. The ITIL (Information Technology Infrastructure Library) framework is the most widely adopted ITSM standard globally, providing guidance across service strategy, design, transition, operation, and continual improvement. Within ITSM, Request Management handles standard service requests — predefined, pre-approved requests such as software installation, access provisioning, or network configuration — separately from incidents or changes, thereby reducing IT workload and improving response times.

2.2 ServiceNow as an Enterprise ITSM Platform

ServiceNow is an industry-leading cloud-based ITSM platform providing a unified system for managing IT services, workflows, and operations. Its Service Catalog module allows organizations to publish a consumer-style catalog of available IT services through which end users can submit structured requests and track their status in real time. Flow Designer enables no-code and low-code workflow automation, while role-based access control ensures data security and appropriate visibility across all user groups.

2.3 Network Request Management Challenges

Research in enterprise IT operations consistently identifies network request management as one of the most process-intensive and error-prone areas in IT service delivery. Organizations without automated request handling experience significantly higher Mean Time to Resolution (MTTR), lower customer satisfaction, and greater risk of non-compliance with security policies. Automation of network request workflows has been shown to reduce MTTR by up to 60%, eliminate manual data entry errors, and substantially improve SLA compliance rates.

3. System Design

3.1 Solution Architecture

The Automated Network Request Management System is built entirely within the ServiceNow enterprise PaaS cloud and is organized around four primary processing layers:

Layer	Description
Presentation Layer	ServiceNow Service Portal for requesters; Native ITSM Workspace for engineers and approvers.
Application & Security Layer	UI Policies and Client Scripts for dynamic form behaviors and input validation; ACLs for role-based data access control.
Automation Engine	Flow Designer orchestrates multi-stage approval routing, conditional branching, SLA tracking, and email notification dispatch.
Storage Layer	Cloud-hosted ServiceNow database tables: sc_req_item (requests), sysapproval_approver (approvals), sc_task (fulfillment tasks), and custom u_network_database table.

3.2 Role-Based Access Control Matrix

Role	Assigned To	Permissions
sn_network_requester	IT Staff, Developers, Business Users	Submit requests; view own submissions only.
sn_network_approver	Line Managers, Security Team Leads	Review and approve/reject assigned requests; add comments.
sn_network_engineer	Network Operations Engineers	View and execute assigned fulfillment tasks; update work notes.
admin	IT Administrators	Full platform access; configuration, reporting, and audit.

3.3 Catalog Item Types

- Network Access Request — granting or modifying employee network access permissions.
- Firewall Rule Change Request — adding, modifying, or removing firewall rules.
- VPN Access Provisioning — setting up VPN credentials and profiles for remote users.
- VLAN Configuration Request — creating or modifying VLAN assignments for network segments.
- IP Address Allocation — requesting static IP assignment for servers or devices.
- Bandwidth Upgrade Request — requesting increased bandwidth allocation for specific links or sites.
- DNS / DHCP Change Request — modifying DNS records or DHCP scope configurations.
- Wireless Network Access Request — provisioning guest or corporate wireless network access.

4. Project Milestones

Milestone 1: Service Catalog & Portal Setup

Establish Service Catalog taxonomy, configure custom roles (sn_network_requester, sn_network_approver, sn_network_engineer), and build the baseline Requester Service Portal with 8 categorized network request items.

Key Deliverables:

- Operational Service Catalog with 8 network request items.
- Custom roles configured with ACLs.
- Service Portal configured for requester access.

Milestone 2: Dynamic Form Development

Build dynamic, intelligent catalog variable forms. Implement UI Policies to conditionally show/hide fields based on user selections. Apply Client Script regex validation to enforce correct IP address formats, CIDR blocks, port numbers, and VLAN identifiers.

Key Deliverables:

- Fully validated dynamic forms for all 8 catalog items.
- UI Policy configurations documented.
- Client Script validation rules active.

Milestone 3: Workflow Automation & Notifications

Configure multi-step approval workflows in Flow Designer routing requests through Line Manager approval and Network Security review. Automate fulfillment task creation, SLA tracking activation, and email notification dispatch at every lifecycle stage.

Key Deliverables:

- Configured approval workflows for all catalog items.
- Automated email notification templates deployed.
- SLA monitoring and escalation rules active.

Milestone 4: Testing & Quality Assurance

Conduct unit testing, integration testing, end-to-end scenario testing, and User Acceptance Testing (UAT) with stakeholder representatives. Log and resolve all critical and high-severity defects prior to production deployment.

Key Deliverables:

- Test plan and test case documentation.
- UAT sign-off from all stakeholder groups.
- Defect log with resolution status.

Milestone 5: Production Deployment

Migrate all configurations from the development instance to production using ServiceNow Update Sets. Conduct post-migration validation, deliver training to all user groups, and establish post-deployment support processes.

Key Deliverables:

- Fully operational system in production.
- Training materials distributed to all user groups.
- Post-deployment support process established.

5. Implementation Details

5.1 ServiceNow Modules Used

Module	Purpose
Service Catalog	Publishing and managing network request catalog items and categories.
Flow Designer	Building automated approval workflows and fulfillment task triggers.
Form Designer / Catalog Variables	Creating dynamic, validated forms for each request type.
UI Policies & Client Scripts	Implementing dynamic form behaviors and field-level validation logic.
Notification Engine	Configuring and dispatching automated email notifications at every stage.
SLA Management	Defining, tracking, and escalating SLAs for each request category.
Assignment Rules	Routing fulfillment tasks to appropriate network engineering groups.
Update Sets	Packaging and migrating configurations between development and production environments.
Reporting & Dashboards	Monitoring request volumes, SLA performance, and team productivity metrics.

5.2 Representative Workflow — Firewall Rule Change Request

- **Trigger:** Request submitted via Service Catalog form.
- **Step 1 — Manager Approval:** Request routed to requester's line manager. Email notification dispatched. SLA timer starts.
- **Step 2 — Network Security Review:** Upon manager approval, request escalated to Network Security team for technical vetting. Conditional routing applied based on risk classification.
- **Step 3 — Fulfillment Task Creation:** Upon security approval, a fulfillment task automatically created and assigned to the Network Operations Group.
- **Step 4 — Task Execution:** Network engineer implements the requested firewall rule change and updates the task record with completion notes.
- **Step 5 — Request Closure:** System automatically closes the parent request and dispatches a completion notification to the requester.
- **Rejection Path:** If rejected at any stage, requester notified with reason. Request record updated with rejection details and an immutable audit log entry created.

6. Testing and Quality Assurance

6.1 Testing Strategy

A comprehensive four-phase testing strategy was employed prior to production deployment: unit testing, integration testing, end-to-end testing, and User Acceptance Testing (UAT). Negative testing was also conducted to validate error handling, security enforcement, and SLA escalation logic.

6.2 Key Test Scenarios

- Successful submission of each catalog item with valid data — confirmed form submission, workflow trigger, and submission notification.
- Submission with missing mandatory fields — confirmed validation errors prevent submission.
- Approval workflow routing — confirmed requests routed to correct approvers at each stage.
- Manager approval followed by security review — confirmed conditional multi-step routing.
- Request rejection at each approval stage — confirmed rejection notifications and audit records.
- Fulfillment task creation and assignment upon approval — confirmed task creation and routing.
- SLA breach simulation — confirmed escalation notifications triggered at 50%, 75%, and 100% thresholds.
- Unauthorized access attempt — confirmed role-based restrictions enforced correctly.
- End-to-end closure — confirmed request and task closure sequence and closure notification delivery.

6.3 UAT Outcomes

User Acceptance Testing was conducted with representatives from IT operations, network engineering, line management, and end-user communities. All critical and high-severity defects identified during UAT were resolved prior to production deployment. Minor enhancement requests were logged in the product backlog for future iterations. UAT sign-off was obtained from all stakeholder groups.

7. Results and Outcomes

7.1 Key Achievements

- Fully operational Service Catalog with 8 categorized network request types accessible through the ServiceNow Service Portal.
- Dynamic, validated forms for all catalog items with conditional field logic, mandatory field enforcement, and user-friendly help text.
- Automated multi-step approval workflows for all request types with conditional routing and email notifications at every stage.
- Fully configured SLA monitoring with automated escalation at 50%, 75%, and 100% breach thresholds.
- Automated fulfillment task creation and routing to network engineering groups, eliminating manual task assignment.
- Complete audit trail for all requests, approval decisions, and fulfillment activities, supporting compliance and reporting requirements.
- Successful production deployment following thorough testing and UAT sign-off by all stakeholder groups.
- Training delivered to all user groups with comprehensive supporting documentation distributed.

7.2 Benefits Realized

- Elimination of email and phone-based request intake, replaced with a structured, trackable catalog process.
- Reduction in incomplete request submissions through mandatory field enforcement and dynamic form guidance.
- Improved request visibility for requesters, approvers, and network teams through real-time status tracking.
- Enhanced compliance through enforced approval chains and immutable audit logging.
- Reduced administrative burden on IT and network teams through workflow automation and automated task creation.
- Improved SLA adherence through automated monitoring and escalation.
- Standardized request data enabling meaningful reporting and trend analysis.

8. Conclusion

8.1 Summary

The Automated Network Request Management System in ServiceNow represents a meaningful advancement in how network service requests are managed within an enterprise. By replacing informal, fragmented request channels with a structured, automated, and ITSM-aligned system, the project delivers measurable improvements in efficiency, compliance, visibility, and user experience across all stakeholder groups. The five-milestone project structure provided a logical pathway from initial catalog design through form development, workflow automation, rigorous testing, and successful production deployment.

8.2 Future Enhancements

- Integration with network management platforms (e.g., Cisco DNA Center, SolarWinds) to enable automated network configuration execution upon task approval.
- Implementation of a self-service knowledge base within the Service Portal to guide users in selecting the correct catalog item.
- Addition of a request analytics dashboard providing real-time visibility into request volumes, SLA performance, and approval cycle times.
- Extension of the catalog to cover additional IT domains such as cloud resource provisioning and server infrastructure requests.
- Chatbot integration using ServiceNow Virtual Agent to enable conversational request submission and real-time status inquiry.

References

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